



JABATAN KEHAKIMAN SYARIAH NEGERI JOHOR

Portal Hakam JKSNJ

Development Progress Report

Kaedah-Kaedah Hakam (Negeri Johor) 2022

Date: 23 April 2026 **Phase:** Phase 2 – Local Backend **Status:** Demo Ready

1. Project Overview

The Portal Hakam JKSNJ is a web-based case management system designed to digitise and streamline the arbitration process (*Hakam*) under the **Kaedah-Kaedah Hakam (Negeri Johor) 2022**. The system manages case registration, session reports, and all 9 statutory forms (Borang 1–9) under Jadual Pertama.

This report documents the development work completed during the current sprint, covering the migration from a browser-local prototype to a fully functional local backend system.

2. Work Completed

2.1 Local Backend Server

A production-capable local server was built using **Node.js + Express.js** with a **SQLite** database. The entire backend lives in a single file (`server.js`) for simplicity and easy maintenance.

Component	Technology	Purpose
HTTP Server	Express.js 4.x	Handles all API requests and serves the frontend
Database	SQLite (better-sqlite3)	Persistent storage — single file, no installation needed
Authentication	express-session + bcryptjs	Session-based login with hashed passwords
File Uploads	multer	Handles PDF/image attachments for Laporan & Borang

2.2 Database Schema

Four tables were designed and implemented:

Table	Key Fields	Notes
users	id, nama, kata (hashed), jawatan, peranan	Roles: admin, pendaftar, hakim
kes	id, data (JSON), created_at, updated_at	Full case object stored as JSON blob
laporan	id, kes_id, data (JSON), created_at, updated_at	Linked to kes; supports file attachment
borang_submissions	id, borang, kes_id, data (JSON), fail, tarikh, user_nama	Tracks all Borang 1–9 submissions

2.3 REST API Endpoints

Method	Route	Access	Description
POST	/api/auth/login	Public	Login, returns session cookie
POST	/api/auth/logout	All	Destroys session
GET	/api/auth/me	All	Session restore on page reload
GET	/api/kes	All roles	List all cases
POST	/api/kes	admin, pendaftar	Create new case
PUT	/api/kes/:id	admin, pendaftar	Update case
DELETE	/api/kes/:id	admin only	Delete case
GET	/api/laporan	All roles	List all reports
POST	/api/laporan	admin, pendaftar	Create report (+ file upload)
PUT	/api/laporan/:id	admin, pendaftar	Update report (+ file upload)
DELETE	/api/laporan/:id	admin only	Delete report
GET	/api/borang	All roles	List borang submissions
POST	/api/borang	All roles	Submit form (+ file upload)
DELETE	/api/borang/:id	admin only	Delete submission record

2.4 Role-Based Access Control

Permissions are enforced **server-side** on every request, not just in the UI.

Action	Admin	Pendaftar	Hakam
View all records	✓	✓	✓
Create / edit Kes	✓	✓	✗
Create / edit Laporan	✓	✓	✗
Submit Borang 1–9	✓	✓	✓
Delete any record	✓	✗	✗

2.5 Frontend Migration (localStorage → API)

The original prototype stored data in `localStorage` (browser-only). All data operations were migrated to call the backend API:

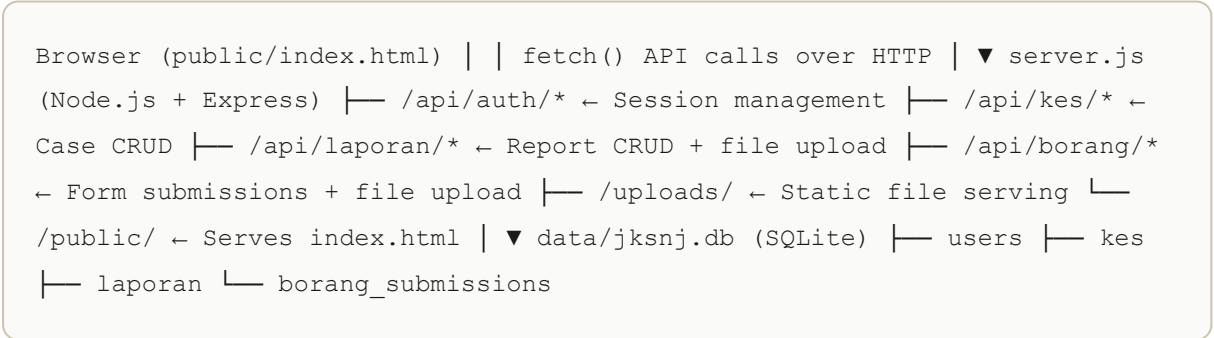
- Login now calls `POST /api/auth/login` with bcrypt password verification
- Session is automatically restored on page refresh via `GET /api/auth/me`
- All CRUD operations (create, read, update, delete) use `fetch()` to the API
- Multiple users on different computers now see the same live data

2.6 File Upload (Laporan Tahkim & Borang)

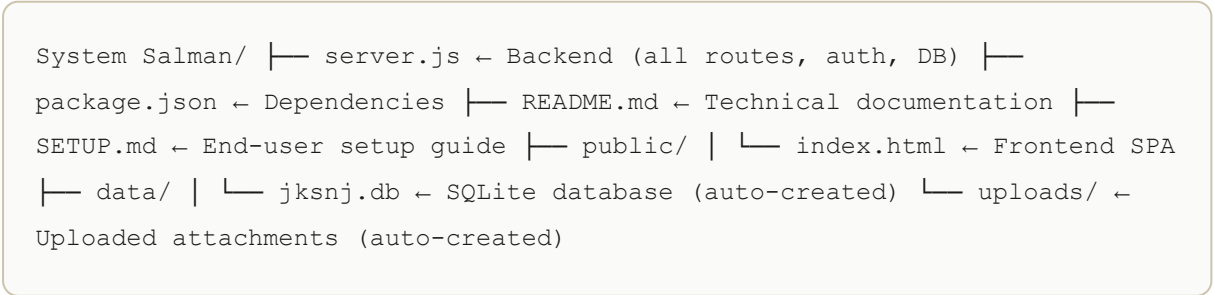
Both the Laporan Tahkim form and all Borang 1–9 forms now support document attachments:

- Accepted formats: PDF, DOC, DOCX, JPG, JPEG, PNG
- Maximum file size: 10 MB per upload
- Files stored in `uploads/` folder on the server
- Attachment link shown in Laporan list view for easy retrieval

3. System Architecture



Project File Structure



4. How to Run

Requirement: Node.js version 18 or above must be installed. Download from nodejs.org.

```
# Step 1 – Install dependencies (first time only)
npm install

# Step 2 – Start the server
npm start

# Step 3 – Open browser
http://localhost:3000
```

Demo Accounts

Username	Password	Role	Access Level
admin	jksnj2025	Pentadbir	Full access including delete
daftar	daftar123	Pendaftar	Create and edit records
hakam1	hakam123	Panel Hakam	View and submit forms only

5. Pending Items

#	Item	Status	Blocker
1	Borang 1–9 input fields — update each form to match exact official form requirements	Blocked	Waiting for official form documents (PDF/Word) from JKSNJ
2	Admin vs user role confirmation — clarify who fills which form	Pending	Confirmation needed from supervisor
3	UI theme — align visual design with official JKSNJ website	Pending	Branding guide / approval needed
4	User management UI — add/change user accounts without editing server.js	Pending	Scope to be confirmed
5	HTTPS / SSL setup for internet or wider LAN deployment	Pending	Required only if moving beyond local network

Note: Items 1 and 2 are dependent on input from the client/supervisor. Development can proceed on items 3–5 independently.

6. Known Limitations (Demo Scope)

- Passwords for demo accounts are visible in `server.js` — should be changed before any real use
- No HTTPS — suitable for local/LAN only; requires SSL certificate for internet deployment
- No automated backup — `data/jksnj.db` and `uploads/` should be backed up manually
- No user management UI — account changes require editing `server.js` and restarting
- Borang 1–9 fields are approximations — pending official form specifications

7. Future Recommendations (If Moving to Production)

- **Cloud deployment** — Host on a server or cloud VM accessible to all JKSNJ branches
- **HTTPS** — Required for any internet-facing deployment (Let's Encrypt for free SSL)
- **Automated backups** — Schedule daily backup of `jksnj.db` to an external drive or cloud storage
- **Audit log** — Record who created/modified/deleted each record (important for court records)
- **MyKad / SSO integration** — Authentication via government identity system
- **Email notifications** — Alert Hakam when approaching 30-day deadline